

Exercise 1 Prolog: Calculating Mathematical Functions

Objective

Learn how to define recursive functions in Prolog.

Statement: We want to implement the function ``Power``: ``Power:  $X^N = X \times X^{N-1}$ ``

Questions: Write the predicate ``power(X, N, R)``

Test with queries

Solution

Puissance

```
puissance(_, 0, 1).
```

```
puissance(X, N, R) :-
```

```
    N > 0,
```

```
    N1 is N - 1,
```

```
    puissance(X, N1, R1),
```

```
    R is X * R1.
```

Test

```
?- puissance(2, 3, R).
```

Résultat :

R = 8.

Exercise 2 Prolog: List Manipulation

Objective: Master:

- Recursion on lists
- Head/tail manipulation
- Standard predicates

Problem Statement: Consider a list of integers.

Questions

1. Write a predicate ``length(L, N)`` that calculates the length of a list.
2. Write a predicate ``sum_of_list(L, S)`` that calculates the sum of the elements.
3. Write a predicate ``max_of_list(L, Max)`` that returns the maximum.
4. Write a predicate ``member(X, L)`` that checks if X belongs to the list.
5. Write a predicate ``concat(L1, L2, L3)`` that concatenates two lists.

Solution

1. Length of a list

```
longueur([], 0).
```

```
longueur([_|T], N) :-
```

```
    longueur(T, N1),
```

```
    N is N1 + 1.
```

Test

```
?- longueur([1,2,3,4], N).
```

Résultat : N = 4.

2. Sum of the elements

```
somme_liste([], 0).
```

```
somme_liste([H|T], S) :-
```

```
    somme_liste(T, S1),
```

```
    S is H + S1.
```

Test

```
?- somme_liste([1,2,3,4], S).
```

Résultat : S = 10.

3. Maximum of a list

max_liste([X], X).

max_liste([H|T], Max) :-

max_liste(T, MaxT),

(H > MaxT -> Max = H ; Max = MaxT).

Testing

?- max_liste([3,7,2,9,5], M).

Résultat : M = 9.

Quick explanation

- We compare the head H with the maximum of the remainder MaxT
- We keep the larger one

4. Membership test

membre(X, [X|_]).

membre(X, [_|T]) :-

membre(X, T).

Testing

?- membre(3, [1,2,3,4]).

Résultat : true.

5. List Concatenation

concat([], L, L).

concat([H|T], L2, [H|R]) :-

concat(T, L2, R).

Test

?- concat([1,2], [3,4], L).

Résultat : L = [1,2,3,4].

Explanation

- Base case:

Concatenating an empty list with L gives L

- Recursive case:

We take the head H of the first list
and concatenate the rest T with L2

Exercise 3 Prolog: Managing an intelligent schedule

Objective

Model a simple scheduling system (inspired by an AI assistant) using Prolog.

Statement We want to represent the following tasks:

Tsk	Duration (h)	Priority
Article	4	haute
Cours	3	haute
Reunion	2	moyenne
correction	5	moyenne
Sport	3	basse

Questions

1. Represent the facts in Prolog
2. Write a rule: important(X) if the priority is high
3. Write a rule: long(X) if the duration > 3
4. Write a rule: priority(X) if the task is both important AND long
5. Write a query to display the priority tasks

Correction

1. Facts

tache(article, 4, haute).
tache(cours, 3, haute).
tache(reunion, 2, moyenne).
tache(correction, 5, moyenne).
tache(sport, 3, basse).

2. Rule : important task

important(X) :-
 tache(X, _, haute).

3. Rule : Long task

longue(X) :-
 tache(X, Duree, _),
 Duree > 3.

4. Rule : priority task

prioritaire(X) :-
 important(X),
 longue(X).

5. Requests

Example : ?- prioritaire(X).

Expected result :

X = article.

Explication :

- article → durée 4 ✓ + priorité haute ✓
- cours → priorité haute mais durée 3 ✗

Exercise 4: Time Optimization with Reclaim.ai

Objective Understand how an AI tool like **Reclaim.ai** can optimize time management and productivity.

Statement: A professor uses Reclaim.ai to organize their week.

They have the following tasks:

Task	Duration	Priority	Deadline
Writing article	4h	High	Friday
Preparing lecture	3h	High	Monday
Lab meeting	2h	Medium	Wednesday
Grading papers	5h	Medium	Sunday
Sports	3h	Low	Flexible

Constraints:

- Work between 9am and 5pm
- **Lunch break:** 12pm–1pm
- **Fixed lab meeting:** Wednesday 10am–12pm
- The system must balance work and well-being

Questions & Solutions

1. Understanding Reclaim.ai's Role in Time Management

Reclaim.ai is an AI-powered scheduling assistant that integrates with your calendar (e.g., Google Calendar) to automatically organize your time.

It helps you:

- Intelligently plan your tasks, routines, and meetings
- Maintain focus and avoid work overload
- Dynamically adapt to changes in your schedule

In short, it acts as a smart personal calendar that continuously optimizes your schedule.

2. Application

Proposal for an optimized weekly schedule

Here is an example of how Reclaim.ai could structure a balanced week:

Monday – Friday

- 8:00 AM – 9:00 AM → Planning + checking emails
- 9:00 AM – 11:00 AM → In-depth work (concentration time protected by AI)
- 11:00 AM – 12:00 PM → Meetings / collaboration
- 12:00 PM – 1:00 PM → Break
- 1:00 PM – 3:00 PM → Project work / research
- 3:00 PM – 4:00 PM → Flexible time slot (rescheduled by AI)
- 4:00 PM – 5:00 PM → Light tasks / administrative tasks

Weekly habits (automatically scheduled by AI):

- Exercise: 3 times per week (Evenings)
- Reading/Learning: 30 minutes per day
- Weekly review: Friday afternoon

Reclaim.ai allows you to:

- Reschedule tasks for meetings
- Preserve dedicated time slots for concentration
- Automatically rebalance workload

3. Analysis

Advantages of AI in this tool

AI provides several key advantages:

- Dynamic scheduling → real-time adjustment based on changing events
- Priority optimization → prioritizes scheduling of important tasks
- Conflict resolution → automatically avoids overlaps
- Time protection → protects periods of concentration from interruptions
- Personalization → learns your habits and preferences

☞ Result: increased productivity with less manual scheduling.

4. Critique

Two limitations of using Reclaim.ai

1. Dependence on accurate input data

- If tasks, priorities, or durations are poorly defined, the AI produces suboptimal schedules.

2. Limited contextual understanding

- The AI may not fully grasp human factors (fatigue, motivation, unforeseen emergencies).

Another limitation:

- Requires trusting the automation, which may be a source of concern for some users.

Conclusion

Reclaim.ai is a powerful tool for automating time management, but it works best when:

- your input data is clear
- and you continue to use human judgment when necessary.